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Periodontitis oral smORFs yield micropeptides that occupy the AChE peripheral site: 22-task UniDL4BioPep scoring, docking, and 100-ns dynamics

Abstract

Periodontitis has been tied to Alzheimer's disease (AD) in clinical and experimental work, yet a peptide-level ligand that could act on a synaptic enzyme is still missing. Here we scored already-processed oral small open reading frames (smORFs) with UniDL4BioPep, docked twelve 7–9-residue peptides into human acetylcholinesterase (AChE, PDB 4EY6), and ran 100-ns molecular dynamics (MD) on apo AChE and three complexes. The healthy-labelled library (11,269,961 sequences) and the periodontitis-labelled library (11,721,988 sequences) were both passed through 22 UniDL4BioPep classifiers at ≥ 0.80 . Only the periodontitis branch was then matched to oral genomic and metaproteomic catalogues. Blood–brain-barrier (BBB) hits numbered 1,095,861 (9.72%) in the healthy library and 1,125,832 (9.60%) in the periodontitis library. Intersection of the periodontitis BBB set with 33,786 unique, catalogue-supported peptides left 3,518 sequences; NTxPred2, mebipred and AnOxPePred reduced that list to twelve explicit peptides. All twelve carry a net positive charge at pH 7.4. Local AutoDock Vina (three runs) gave best poses between -8.25 and -9.60 kcal/mol. FLLHTTR, YLSLLQR and LLHPLRL contact the peripheral anionic site (PAS). Over 100 ns the FLLHTTR and YLSLLQR complexes stay more compact than apo AChE (backbone RMSD 0.1640 and 0.1625 nm versus 0.1897 nm). FLLHTTR keeps a dense hydrogen-bond net (7.03 ± 1.28); only YLSLLQR shrinks solvent-accessible surface area. These calculations outline a possible route in which oral micropeptides sit on the same PAS that accelerates amyloid- β (A β) assembly.

Keywords: Alzheimer's disease; *Porphyromonas gingivalis*; periodontitis; smORF; micropeptide; acetylcholinesterase; peripheral anionic site; UniDL4BioPep; molecular docking; molecular dynamics

Introduction

Alzheimer's disease (AD) combines amyloid deposition, tau pathology, synaptic failure, immune activation and vascular injury over a long preclinical course [1]. Sequential cleavage of APP by β - and γ -secretases releases A β 40 and A β 42; soluble oligomers injure synapses; familial APP/PSEN mutations change peptide length and amount [2]. Amyloid load does not explain the spatial or clinical heterogeneity of the disease, so peripheral inflammatory exposures have been examined as possible modifiers of vulnerability rather than as single sufficient causes.

Loss of basal-forebrain acetylcholine accounts for a large fraction of the cognitive picture, which is why AChE inhibitors remain in routine use [3]. Catalysis is only part of the enzyme's role. AChE accelerates A β fibril growth through the peripheral anionic site (PAS), and AChE–A β particles are more toxic than free peptide [4]. A short hydrophobic PAS motif is sufficient for that chaperone effect [5]. PAS-directed small molecules can block AChE-induced aggregation in biochemical assays [19]. The same protein surface therefore joins cholinergic failure to amyloid deposition.

Chronic periodontitis maintains a low-grade inflammatory load at a disrupted mucosal barrier and allows microbial products into blood [6]. Oral activity is species- and site-specific, so 16S abundance cannot stand in for a molecular ligand [7]. Meta-analyses report associations between periodontal disease and cognitive disorders, although effect sizes move with case definitions [10]. In an AD cohort, periodontitis tracked later decline [11]. A two-sample Mendelian randomization analysis did not support a genetic causal effect of periodontal disease

on AD [14]. Epidemiology therefore motivates a molecular search; it does not identify the ligand.

Gingipains and outer-membrane vesicles of *Porphyromonas gingivalis* supply one mapped virulence pair [8–9]. The organism and gingipains have been reported in AD brains [12], and repeated oral infection in mice produces neuroinflammation and A β -related changes [13]. Those observations justify looking at oral products. They do not, by themselves, name a peptide that occupies AChE.

Microbiome smORFs encode a large pool of uncharted small proteins [15–16]. Mining of human microbiomes for peptide antibiotics first scored millions of translated open reading frames and then applied experimental filters [31]. UniDL4BioPep supplies more than twenty binary bioactivity heads on ESM-2 embeddings [23]. That order—predict, then match to catalogues—is the order used here. Classifier scores do not answer the structural question of whether a 7–9-aa periodontitis peptide can occupy the A β -binding PAS.

Accelerated MD places A β on the AChE surface and treats the enzyme as a nucleation centre [17]. A 1- μ s PAS-centred AChE–A β trajectory remains bound, with the main residence at residues 344–361 [18]. PDB 4EY6 gives a 2.40 Å human AChE frame for docking [20]. The aromatic gorge that joins the catalytic triad to the PAS was mapped on *Torpedo* AChE [36]. What remains missing is a peptide-level ligand drawn from oral smORFs and tested on that same PAS.

Here we scored already-processed oral smORF libraries from PRJNA678453 with 22 UniDL4BioPep heads, compared healthy and periodontitis hit rates, matched the periodontitis branch to oral genomic and metaproteomic catalogues, and reduced the list with NTxPred2, mebipred and AnOxPePred. Twelve 7–9-aa peptides were docked into human AChE. Three complexes, together with apo enzyme, were simulated for 100 ns to test whether the peptides remain at the PAS without unfolding the fold.

Materials and methods

Study design

The work is computational. No new patients, specimens, sequencing runs or wet assays were added. Healthy and periodontitis tags are library labels; they are not peptide-level clinical diagnoses. Docking used local three-run AutoDock Vina poses. MD used 100-ns GROMACS trajectories of apo AChE and three peptide complexes.

Source libraries

Translated smORFs of 4–50 aa were taken as already-processed peptide strings. We did not reassemble reads or call genes de novo. The public source of the paired oral metagenomes and metatranscriptomes is PRJNA678453 [7]. A derived MGnify third-party assembly, PRJEB65451 (metaSPAdes v3.15.3), exists for that BioProject; it is not a second clinical cohort. Library sizes were 11,269,961 healthy-labelled and 11,721,988 periodontitis-labelled sequences.

UniDL4BioPep (22 tasks)

UniDL4BioPep was applied first to both full libraries, following the predict-then-filter order used when mining microbiome peptide antibiotics [23,31]. Du et al. freeze ESM-2 (esm2_t6_8M_UR50D; 6 layers, 8 million parameters, 320-dimensional residue states) and average residue embeddings so that peptides of any length collapse to one fixed vector. A convolutional classifier is then trained, per bioactivity, on that vector; the same backbone is reused across datasets rather than designing a new network for each endpoint. In the original report the architecture beat the previous state of the art on 15 of 20 binary peptide tasks. We did not retrain the networks. We scored 22 published heads: ACE inhibitory, DPP-IV inhibitory, Bitter, Umami, Antimicrobial, Antimalarial (alternative), Antimalarial (main), Quorum sensing, Anticancer (main), Anticancer (alternative), Anti-MRSA, TTCA, BBB (BBP), Anti-

parasitic (APP), NeuroPred, Antibacterial, Antifungal, Antiviral, Toxicity, Antioxidant FRS, Allergenicity, and cell-penetrating peptide (CPP). Every head used a score cut of ≥ 0.80 . “BBB-high” is that operational cut on the BBB head, not measured transcytosis. Other BBB peptide tools (for example Augur) are trained on different positives and encodings, so their published AUCs do not transfer to these 4–50-aa oral strings [35].

Catalogue matching (periodontitis branch only)

After scoring, only periodontitis-labelled sequences were exact-matched to oral genomic and metaproteomic resources and collapsed to unique peptides. HOMD and eHOMD supply curated aerodigestive genomes [21,32]. Salivary metaproteomes record peptides inside their own false-discovery framework [22]. Further oral metaproteomic sets add sequence observations from other clinical contexts [33–34]. A match supports prior observation of the string; it does not prove expression in PRJNA678453 samples. The periodontitis library yielded 33,786 unique catalogue-supported peptides. Intersection with the 1,125,832 periodontitis BBB (BBP) hits recovered 3,518 sequences (3,446 of 5–30 aa; 72 of 31–50 aa). The healthy library was left at the 22-task score tables and was not dereplicated.

NTxPred2

NTxPred2 trains separate peptide and protein models because a single neurotoxin classifier does not transfer between length classes [24]. The peptide set has 877 neurotoxic and 877 non-toxic sequences; the protein set has 775 of each. Composition and binary-profile machine-learning models reach AUC 0.97 (peptides) and 0.85 (proteins). Fine-tuning ESM2-t30 on the peptide set raises independent-set AUC to 0.98 (MCC 0.90). The public server uses ESM2-t30 for 7–50-aa inputs (default probability 0.5) and an extra-trees model for sequences ≥ 51 aa. We used the peptide (ESM2-t30) mode on 7–50-aa members of the 3,518-set. A positive call is a classifier label, not a neuronal assay.

mebipred

mebipred predicts metal-binding from sequence alone, without alignments, so it can be applied to short translated fragments [25]. Inputs are 219 sequence features: amino-acid composition, physicochemical descriptors, and counts of metal-binding 5-mers. A first feed-forward net (two hidden layers of 219 ReLU units, dropout 0.2, RMSprop) separates metal-binding from non-binding proteins; a second tier names 11 ions. The authors report >80% accuracy for ligand presence and an area under the precision–recall curve of 0.91. Default ion probability is 0.5; raising it to 0.9 increases precision at some cost in recall. We kept Cu-, Fe- and Zn-related scores at 0.50. The output is not a measured K_d and does not assign a coordination geometry.

AnOxPePred

AnOxPePred is a one-dimensional CNN with two heads, trained to score free-radical scavenging (FRS) and metal chelation (CHEL) from one-hot peptide sequences [26]. Positives were taken from BIOPEP-UWM and from antioxidant-peptide papers; each sequence is labelled FRS, chelator, or both. Reported FRS performance exceeds a k-NN baseline; the chelator head is weaker because that training subset is small. Both heads emit a score in [0, 1]. Serial cuts here were $CHEL \geq 0.25$, then $CHEL \geq 0.25$ with $FRS < 0.50$, then $CHEL \geq 0.25$ with $FRS < 0.45$. Concordance among UniDL4BioPep, NTxPred2, mebipred and AnOxPePred is a filter stack, not independent experimental replication.

Physicochemical descriptors

For the twelve unique 7–9-aa strings we recalculated length, histidine, cysteine and Arg+Lys counts, average molecular weight, isoelectric point, net charge at pH 7.4, Kyte–Doolittle GRAVY, Ikai aliphatic index, hydrophobic residue fraction (A, I, L, M, F, V, W, Y),

and Boman index. Scales were applied to the amino-acid strings only; no experimental HPLC or CD was performed.

Molecular docking

Human recombinant AChE (PDB 4EY6, 2.40 Å) was stripped of galantamine and crystal waters, chain breaks were repaired, and protonation was set at pH 7.4 [20]. The twelve ligands ALLLHRC, FCLHLQLR, FLLHTTR, HLLTLKKHV, HLPLLHRCC, HVLLLRQCA, LLHLPKRTT, LLHPLRC, LLHPLRL, WLLVHLKK, YHHLLCRR and YLSLLQR were docked with AutoDock Vina, exhaustiveness 32 [27–28]. The grid was centred on the PAS (Tyr72, Asp74, Thr75, Leu76, Trp286, His287, Tyr341) and covered the gorge neck (Phe295), the choline subsite (Trp86, Glu202, Tyr337) and the catalytic triad (Ser203, His447, Glu334). Each ligand was run three times. We report best-run affinity, three-run mean $\pm$ SD, hydrogen-bond count and PAS contact from the single best pose. Vina scores rank poses; they are not experimental free energies.

Molecular dynamics

Four explicit-solvent systems were built in GROMACS with Amber99SB-ILDN and TIP3P water at 0.15 M NaCl [29–30]: apo AChE (chain A) and the ALLLHRC, FLLHTTR and YLSLLQR complexes. Each box was triclinic with a 1.0 nm solute-to-wall buffer. Equilibration was 2,000 steps of steepest descent, 1.0 ns restrained NVT to 300 K, 1.0 ns restrained NPT, and 1.0 ns free NPT. Production was 100 ns (dt = 2.0 fs) at 300 K and 1.0 bar with LINCS, 1.2 nm cut-offs and particle-mesh Ewald. Frames were stored every 20 ps.

Metrics matched Figures 5–7: C α RMSD, per-residue RMSF, SASA, R_g, DSSP occupancy, and intermolecular hydrogen bonds (gmx hbond; donor–acceptor $\leq$ 3.0 Å). Peptide self-fit RMSD and persistent contacts (7.0 Å) were stored as extras. Means $\pm$ SD use the

last 20 ns (80–100 ns). The design follows the AChE–A β MD logic of Atanasova et al., at 100 ns rather than 1 μ s [18].

Results

Twenty-two UniDL4BioPep tasks on both libraries

Figure 1 summarises the cascade. Both libraries were scored at ≥ 0.80 on all 22 heads (Tables 1 and 2). Hit rates sit close to each other. Antimicrobial called 10,302,093/11,721,988 periodontitis sequences (87.89%) and 9,882,657/11,269,961 healthy sequences (87.69%). BBB (BBP) called 1,125,832 (9.60%) versus 1,095,861 (9.72%). Anti-parasitic (APP) and quorum sensing were the next largest heads; DPP-IV inhibitory was the smallest in both libraries. Labels overlap, so one peptide can sit in several rows. Because the two BBB rates differ by only 0.12 percentage points, BBB-high is not a periodontitis-specific stamp. The two libraries are close on most of the 22 heads (Table 3). Periodontitis exceeds healthy by ≥ 1 percentage point only for quorum sensing (+1.39). Healthy exceeds periodontitis by ≥ 1 point for antifungal (−2.50), anti-parasitic (−2.36), antioxidant FRS (−1.95), antiviral (−1.89), antibacterial (−1.42), CPP (−1.24), anticancer-main (−1.22) and umami (−1.00). Antimicrobial is essentially tied (87.89% vs 87.69%). These are library-wide score rates, not sequence-level differential peptides: we do not have a row-wise overlap map, so we cannot say that the twelve docking sequences are absent from the healthy library. Downstream docking used the periodontitis, catalogue-matched branch only.

Table 1. UniDL4BioPep counts for the periodontitis-labelled library (11,721,988 smORFs; ≥ 0.80).

No.	Task	n	%
1	ACE inhibitory	1,236,442	10.55
2	DPP-IV inhibitory	139,056	1.19
3	Bitter	1,831,185	15.62
4	Umami	3,100,811	26.45
5	Antimicrobial	10,302,093	87.89

No.	Task	n	%
6	Antimalarial (alternative)	695,608	5.93
7	Antimalarial (main)	2,010,724	17.15
8	Quorum sensing	4,491,507	38.32
9	Anticancer (main)	2,357,718	20.11
10	Anticancer (alternative)	2,015,652	17.20
11	Anti-MRSA	843,977	7.20
12	TTCA	2,666,759	22.75
13	BBB (BBP)	1,125,832	9.60
14	Anti-parasitic (APP)	5,462,493	46.60
15	NeuroPred	1,714,373	14.63
16	Antibacterial	2,597,877	22.16
17	Antifungal	2,960,118	25.25
18	Antiviral	3,275,203	27.94
19	Toxicity	1,714,299	14.62
20	Antioxidant FRS	2,521,106	21.51
21	Allergenicity	1,713,798	14.62
22	Cell-penetrating peptide (CPP)	925,627	7.90

Table 2. UniDL4BioPep counts for the healthy-labelled library (11,269,961 smORFs; ≥ 0.80).

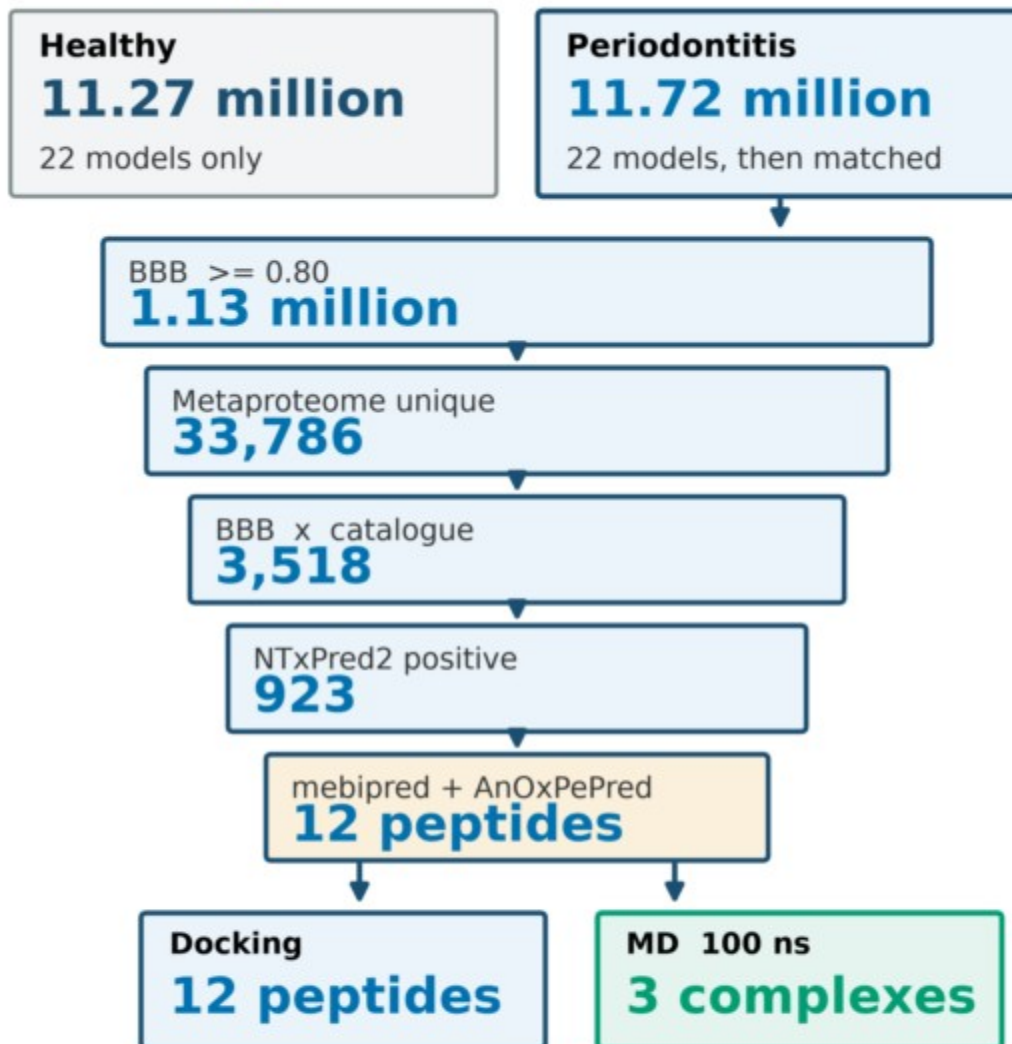
No.	Task	n	%
1	ACE inhibitory	1,237,451	10.98
2	DPP-IV inhibitory	131,426	1.17
3	Bitter	1,840,368	16.33
4	Umami	3,094,287	27.46
5	Antimicrobial	9,882,657	87.69
6	Antimalarial (alternative)	703,632	6.24
7	Antimalarial (main)	1,954,667	17.34
8	Quorum sensing	4,161,825	36.93
9	Anticancer (main)	2,404,084	21.33
10	Anticancer (alternative)	1,979,643	17.57
11	Anti-MRSA	769,955	6.83
12	TTCA	2,618,849	23.24
13	BBB (BBP)	1,095,861	9.72
14	Anti-parasitic (APP)	5,517,278	48.96
15	NeuroPred	1,690,436	15.00

No.	Task	n	%
16	Antibacterial	2,658,234	23.59
17	Antifungal	3,128,057	27.76
18	Antiviral	3,362,295	29.83
19	Toxicity	1,725,268	15.31
20	Antioxidant FRS	2,643,538	23.46
21	Allergenicity	1,635,019	14.51
22	Cell-penetrating peptide (CPP)	1,029,770	9.14

Table 3. Score-rate difference between libraries (periodontitis % minus healthy %). BBB is shown for reference.

Task	Healthy %	Periodontitis %	Δ pp
Quorum sensing	36.93	38.32	+1.39
Umami	27.46	26.45	-1.00
Anticancer (main)	21.33	20.11	-1.22
Cell-penetrating peptide (CPP)	9.14	7.90	-1.24
Antibacterial	23.59	22.16	-1.42
Antiviral	29.83	27.94	-1.89
Antioxidant FRS	23.46	21.51	-1.95
Anti-parasitic (APP)	48.96	46.60	-2.36
Antifungal	27.76	25.25	-2.50
BBB (BBP)	9.72	9.60	-0.12

Screening cascade



Healthy library is scored only and is not proteome-matched.

Funnel numbers are classifier outputs on the periodontitis branch.

Figure 1. Screening cascade. UniDL4BioPep (22 tasks) was applied to both libraries. Catalogue matching and later filters were applied only to the periodontitis branch, ending in twelve 7–9-aa peptides for docking and three complexes for 100-ns MD.

Periodontitis funnel to twelve sequences

Catalogue matching of the periodontitis library kept 33,786 unique peptides. Intersection with BBB-high gave 3,518 sequences. NTxPred2 scored 3,299/3,518 (93.77%) and called 923/3,299 (27.98%) positive. Later cuts left 111 mebipred-positive peptides, 15 with $\text{CHEL} \geq 0.25$, 12 with $\text{CHEL} \geq 0.25$ and $\text{FRS} < 0.50$, and 8 with the stricter $\text{FRS} < 0.45$ (Table 4). All 923 NTxPred2-positive peptides were ≤ 30 aa, so the metal/CHEL/FRS steps kept short peptides only.

Table 4. Periodontitis branch after UniDL4BioPep scoring.

Stage	Rule	n	Denominator
Periodontitis smORFs	4–50 aa	11,721,988	Library
BBB (BBP)	score ≥ 0.80	1,125,832	11,721,988
Catalogue-supported unique peptides	Exact match	33,786	11,721,988
BBB-high $\cap$ catalogue-supported	Intersection	3,518	$1,125,832 \cap 33,786$
Short (5–30 aa)	Length	3,446	3,518
Long (31–50 aa)	Length	72	3,518
NTxPred2 scored	7–50 aa	3,299	3,518
NTxPred2-positive	Model label	923	3,299
mebipred-positive	≥ 0.50	111	—
CHEL-priority	$\text{CHEL} \geq 0.25$	15	111
Main set	$\text{CHEL} \geq 0.25$ and $\text{FRS} < 0.50$	12	111
Stricter subset	$\text{CHEL} \geq 0.25$ and $\text{FRS} < 0.45$	8	—

Physicochemical profile of the twelve peptides

These descriptors characterise the twelve docking ligands. They are not a comparison of the healthy and periodontitis libraries.

The twelve strings are unique 7–9-aa peptides of standard residues (Table 5). Eleven contain histidine; six contain cysteine; every sequence has at least one Arg or Lys. Molecular weights fall between 825.03 and 1,097.30 Da. Isoelectric points are alkaline (8.28–11.54). Net charge at pH 7.4 is positive for all twelve (0.85–2.08). GRAVY is positive for ten sequences, in

line with leucine-rich cores; LLHLPKRTT (−0.36) and YHHLLCRR (−0.95) are the two hydrophilic exceptions. Aliphatic indices run from 97.5 (YHHLLCRR) to 222.9 (LLHPLRL). YLSLLQR is the only peptide without histidine. These numbers describe composition; they are not HPLC or CD measurements.

Table 5. Composition and calculated physicochemical descriptors of the twelve 7–9-aa peptides.

Peptide	aa	MW (Da)	pI	z (pH 7.4)	GRAVY	AI	His	Cys	R+K
ALLHRC	7	825.03	9.00	0.92	1.14	181.4	1	1	1
FCLHLQLR	8	1029.26	9.00	0.92	0.69	146.2	1	1	1
FLLHTTR	7	887.04	11.09	1.04	0.19	111.4	1	0	1
HLLTLKKHV	9	1088.35	10.63	2.08	0.08	162.2	2	0	2
HLPLLHRCC	9	1091.35	8.28	0.85	0.43	130.0	2	2	1
HVLLLQCA	9	1052.30	9.00	0.92	0.97	173.3	1	1	1
LLHLPKRTT	9	1078.32	11.54	2.04	−0.36	130.0	1	0	2
LLHPLRC	7	851.07	9.00	0.92	0.66	167.1	1	1	1
LLHPLRL	7	861.09	11.09	1.04	0.84	222.9	1	0	1
WLLVHLKK	8	1036.32	10.63	2.04	0.46	182.5	1	0	2
YHHLLCRR	8	1097.30	9.91	1.96	−0.95	97.5	2	1	2
YLSLLQR	7	892.06	9.89	0.99	0.04	167.1	0	0	1

Docking to human AChE

All twelve ligands produced favourable Vina scores (Table 6, Figure 2). Best-run values ran from −8.25 to −9.60 kcal/mol; three-run means ran from -8.07 ± 0.16 to -9.44 ± 0.09 kcal/mol. Best-pose order was FLLHTTR (−9.60), YLSLLQR (−9.49), ALLLHRC (−9.29). Mean order put YLSLLQR first (-9.44 ± 0.09) and ALLLHRC second (-9.18 ± 0.11).

FLLHTTR has the strongest single pose and the largest SD (-8.77 ± 1.41). Best poses made 3–10 hydrogen bonds (mean length 2.83–3.28 Å; Figures 3 and 4).

Table 6. Three-run AutoDock Vina scores against human AChE (PDB 4EY6).

Peptide	H-bonds	Best	Mean $\pm$ SD (n=3)	PAS	Principal contacts
ALLLHRC	3	-9.29	-9.18 ± 0.11	No	Ser125, Ser203, Tyr124
FCLHLQLR	7	-9.27	-8.96 ± 0.48	Yes	Ser203, Thr75, Tyr341
FLLHTTR	8	-9.60	-8.77 ± 1.41	Yes	Asp74, Tyr72, His287
HLLTLKKHV	6	-8.88	-8.69 ± 0.20	Yes	Tyr72, Phe346
HLPLLHRCC	4	-8.35	-8.28 ± 0.07	No	Ser125, Tyr124, Tyr337
HVLLLRQCA	4	-8.25	-8.07 ± 0.16	Yes	Thr75
LLHLPKRTT	3	-9.01	-8.89 ± 0.16	Adjacent	Ser203, Val340
LLHPLRC	4	-8.91	-8.78 ± 0.11	No	Ser125, Ser293
LLHPLRL	10	-8.94	-8.91 ± 0.05	Yes	Trp286, Tyr341, His447
WLLVHLKK	4	-8.94	-8.64 ± 0.26	No	Asn283, Gln279
YHHLLCRR	7	-9.03	-8.62 ± 0.43	No	Trp86, Ser203
YLSLLQR	7	-9.49	-9.44 ± 0.09	Yes	Tyr72, Thr75, Glu202

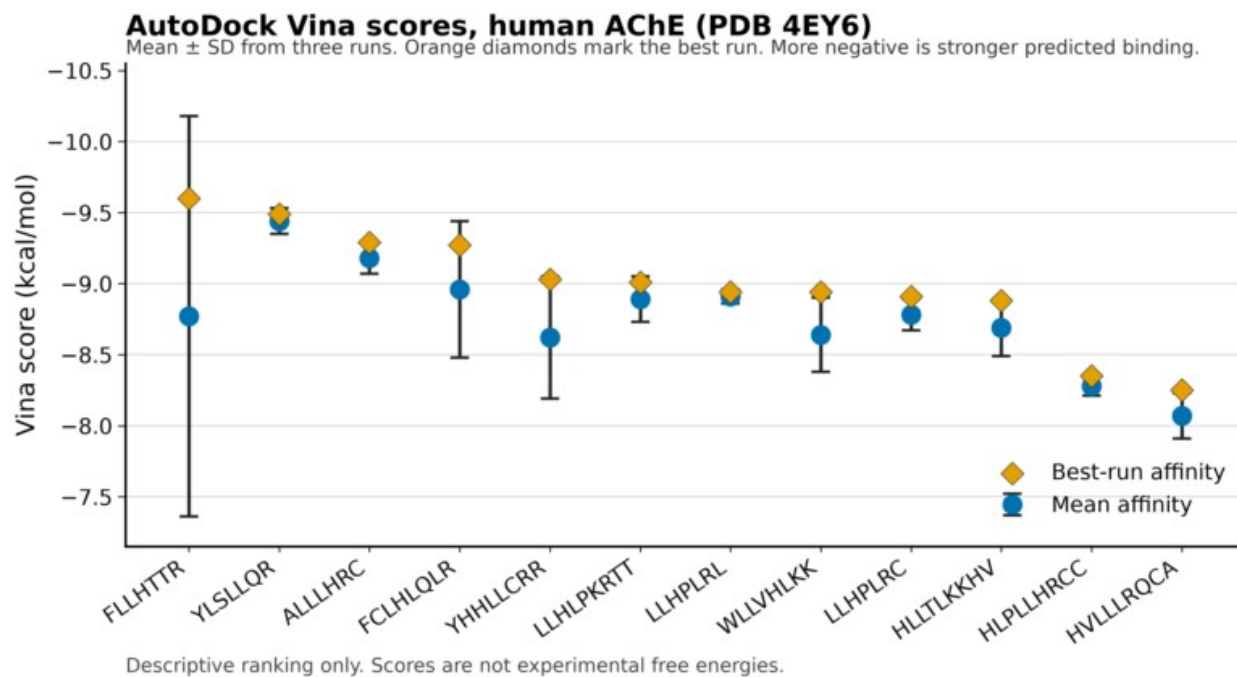


Figure 2. Three-run Vina scores against human AChE (PDB 4EY6). Blue circles, mean; whiskers, SD; orange diamonds, best run. Order follows best-run rank.

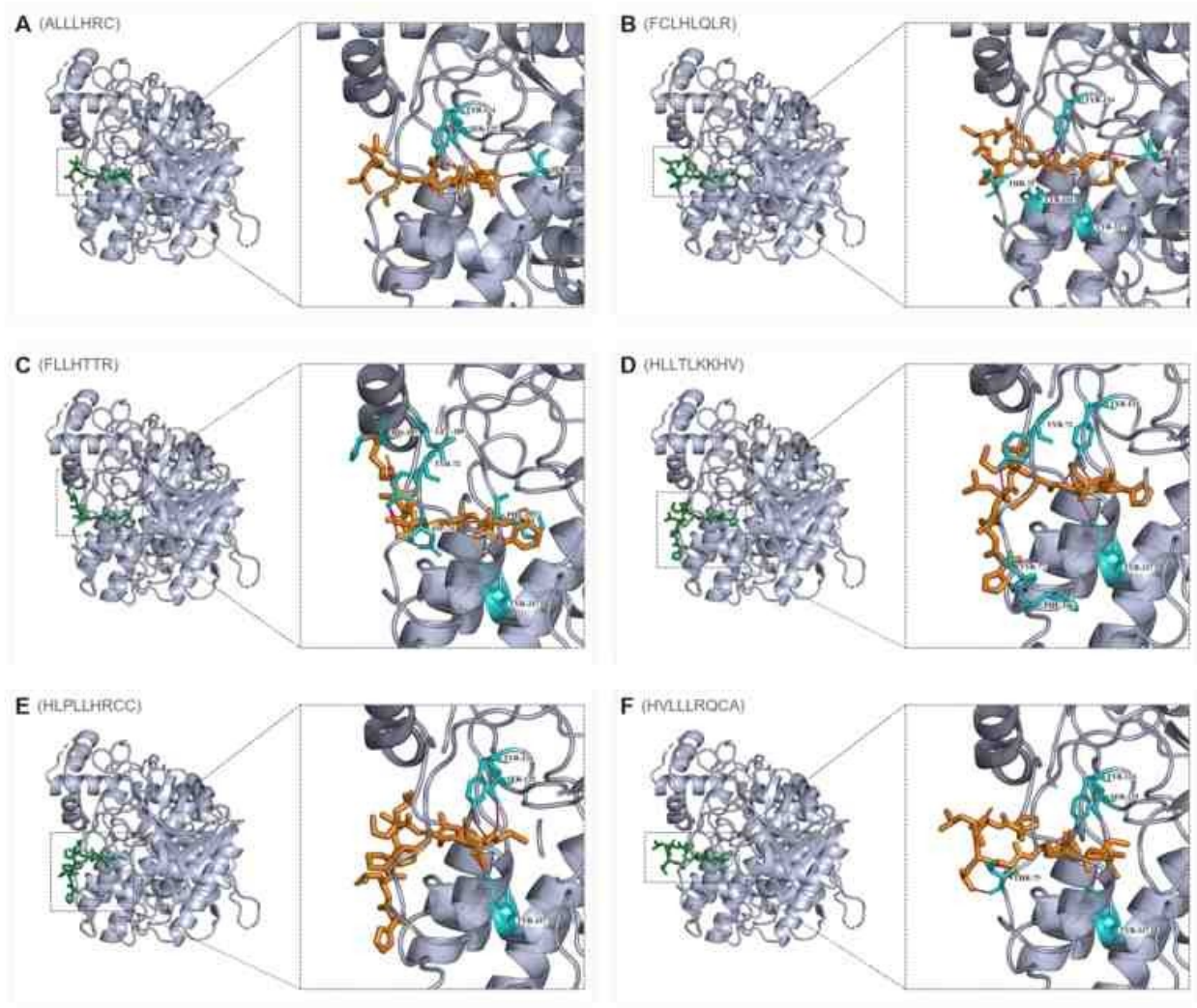


Figure 3. Best docking poses, peptides 1–6 (A–F). Peptide, orange; contacting residues, cyan. FLLHTTR (C) is the densest PAS pose.

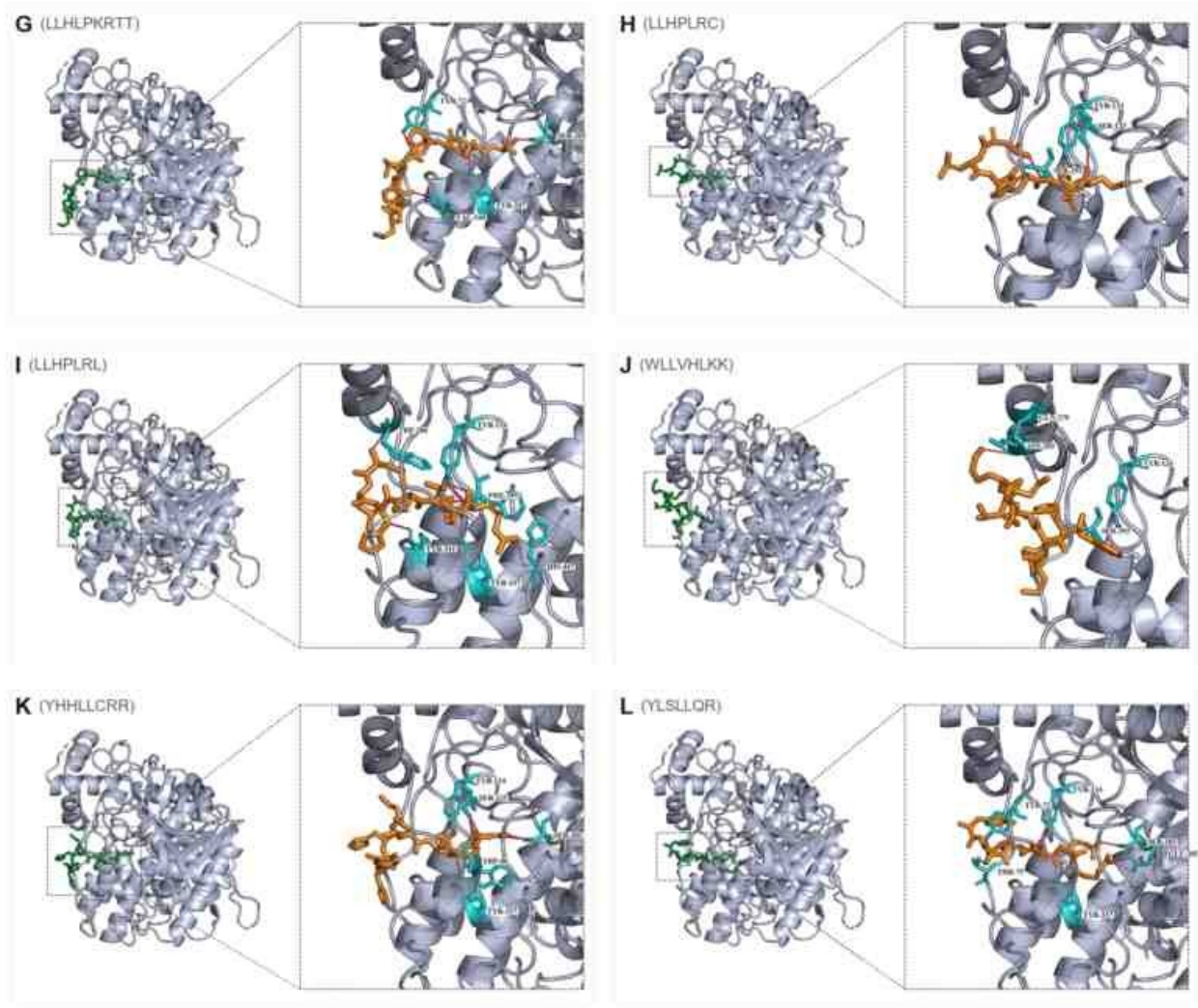


Figure 4. Best docking poses, peptides 7–12 (G–L). LLHPLRL (I) spans PAS Trp286/Tyr341 to catalytic His447. YLSLLQR (L) bridges PAS and the gorge mouth.

PAS contacts in the best pose were seen for FLLHTTR (Figure 3C), YLSLLQR (Figure 4L), FCLHLQLR, HVLLLRQCA, HLLTLKKHV and LLHPLRL (Figure 4I). ALLLHRC binds catalytic Ser203 with the shortest mean hydrogen bond (2.83 Å) rather than the outer PAS aromatics (Figure 3A). Three-run means separate ligands that stay strong across runs (YLSLLQR, ALLLHRC, LLHPLRL) from ligands whose best pose is better than the run average (FLLHTTR, FCLHLQLR, YHLLCRR).

100-ns dynamics of apo AChE and three complexes

Production runs finished for apo AChE and the ALLLHRC, FLLHTTR and YLSLLQR complexes (Table 7, Figures 5–7). Each six-panel figure compares apo with one complex: RMSD (A), RMSF (B), SASA (C), Rg (D), last-20-ns DSSP (E) and intermolecular hydrogen bonds (F).

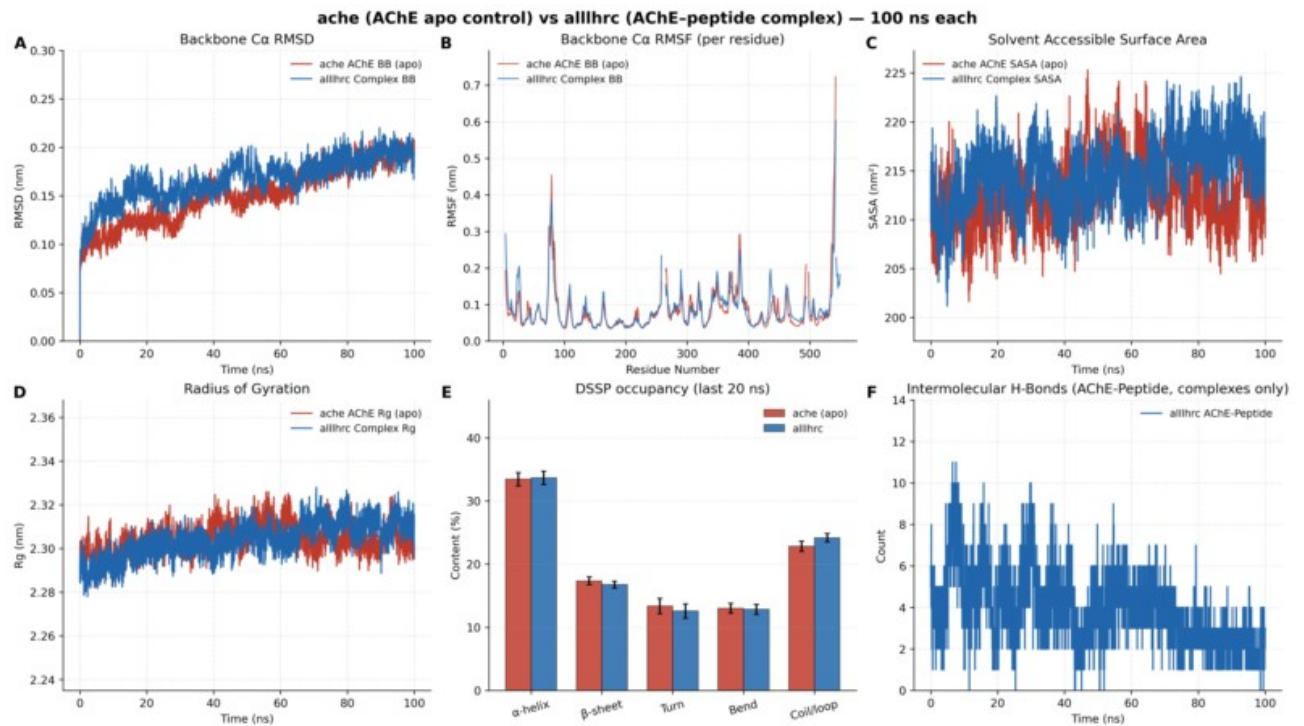


Figure 5. Apo AChE versus AChE–ALLHRC. Panels A–F match Table 7. Complex RMSD (A) tracks apo; hydrogen bonds (F) fall from early occupancy to about two in the last 20 ns.

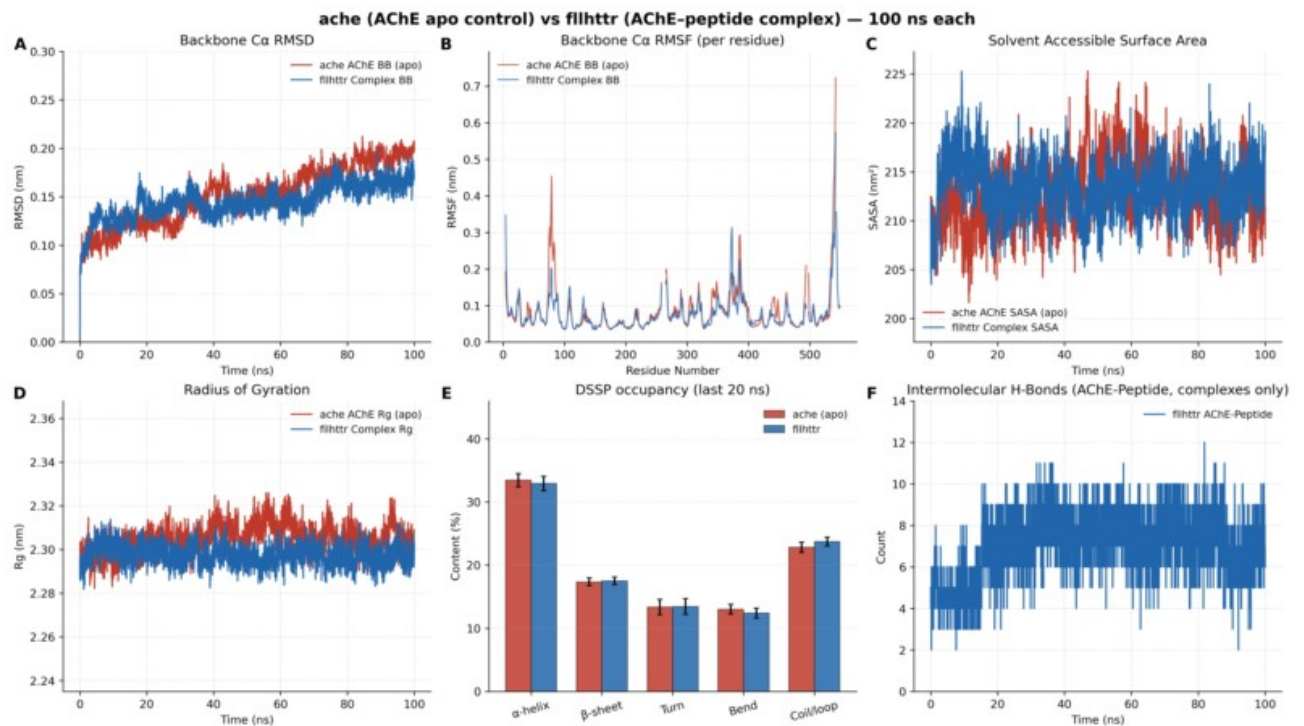


Figure 6. Apo AChE versus AChE–FLLHTTR. After ~50 ns, complex RMSD (A) lies below apo. Hydrogen-bond counts (F) stay in the 6–10 range for the full 100 ns.

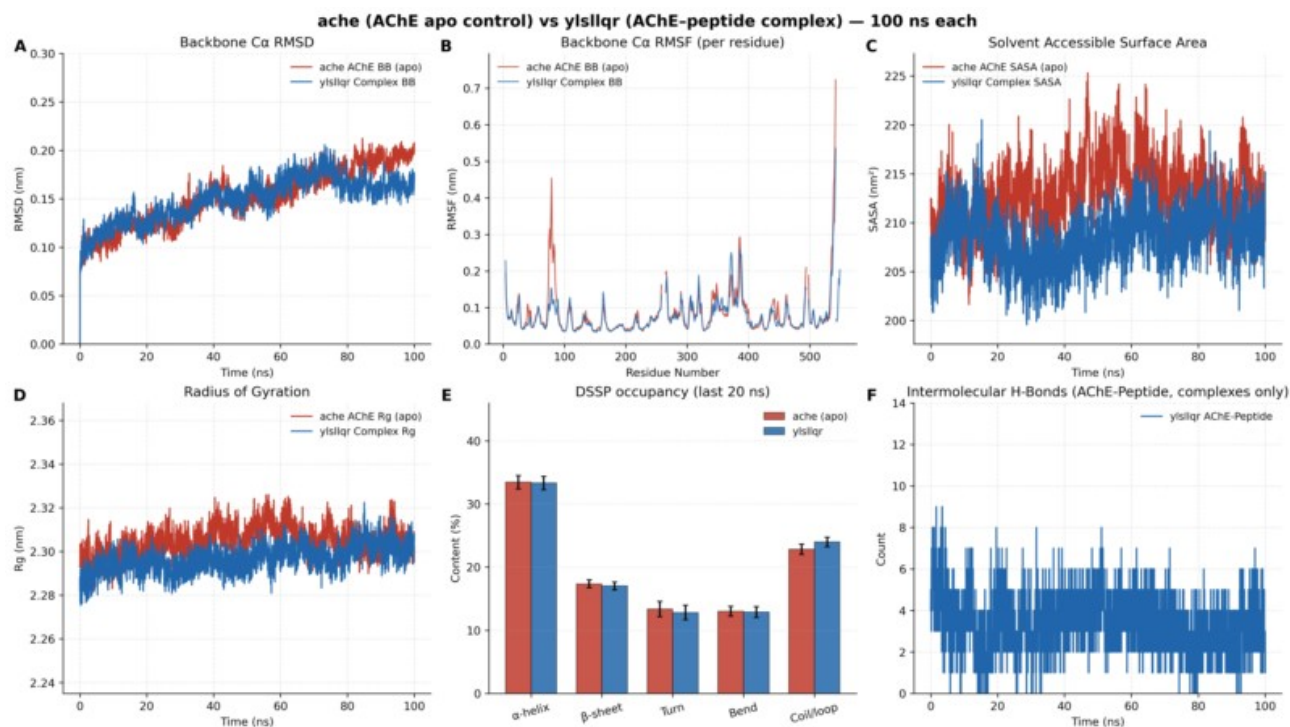


Figure 7. Apo AChE versus AChE-YLSLLQR. Late RMSD (A) is below apo. SASA (C) is the only complex that contracts relative to apo.

Table 7. Last-20-ns metrics (mean $\pm$ SD) for apo AChE and three complexes.

Metric	apo AChE	ALLLHRC	FLLHTTR	YLSLLQR
C α RMSD (nm)	0.1897 $\pm$ 0.0090	0.1916 $\pm$ 0.0092	0.1640 $\pm$ 0.0080	0.1625 $\pm$ 0.0078
Peptide self-fit RMSD (nm)	—	0.2518 $\pm$ 0.0136	0.1752 $\pm$ 0.0111	0.0911 $\pm$ 0.0098
RMSF mean (nm)	0.0835 $\pm$ 0.0659	0.0876 $\pm$ 0.0581	0.0778 $\pm$ 0.0504	0.0771 $\pm$ 0.0498
SASA (nm ²)	212.25 $\pm$ 2.89	217.47 $\pm$ 2.49	213.88 $\pm$ 2.36	209.71 $\pm$ 2.35
Rg (nm)	2.3045 $\pm$ 0.0056	2.3107 $\pm$ 0.0052	2.2967 $\pm$ 0.0047	2.3028 $\pm$ 0.0051
Intermolecular H-bonds	—	2.19 $\pm$ 0.80	7.03 $\pm$ 1.28	2.93 $\pm$ 1.14
Persistent contact pairs	—	7	7	7
DSSP α -helix / β -sheet (%)	33.44 / 17.35	33.66 / 16.76	32.92 / 17.52	33.31 / 17.02

Apo RMSD levels near 0.19 nm (Figures 5A–7A). ALLLHRC follows that control (complex 0.1916 nm). FLLHTTR and YLSLLQR drop below apo after 50–70 ns (0.1640 and 0.1625 nm), which reads as rigidification, not unfolding. Peptide self-fit RMSD is highest for ALLLHRC (0.2518 nm) and lowest for YLSLLQR (0.0911 nm). Catalytic-core RMSF stays

low; mean RMSF is below apo for FLLHTTR and YLSLLQR. R_g remains 2.30–2.31 nm. SASA rises for ALLLHRC (217.47 nm²), sits near apo for FLLHTTR (213.88 nm²), and falls only for YLSLLQR (209.71 nm²; Figure 7C). Hydrogen-bond histories differ: ALLLHRC decays to 2.19 ± 0.80 ; FLLHTTR holds 7.03 ± 1.28 for the whole run (Figure 6F); YLSLLQR averages 2.93 ± 1.14 . Helix (~33%) and sheet (~17%) overlay the apo bars. Each complex keeps seven persistent contact pairs. Centre-of-mass RDF peaks lie at 1.22 nm (ALLLHRC), 1.80 nm (FLLHTTR) and 1.62 nm (YLSLLQR), i.e. surface residence rather than bulk solvent.

Discussion

Principal findings

Twenty-two UniDL4BioPep heads scored 11.27 million healthy-labelled and 11.72 million periodontitis-labelled oral smORFs at ≥ 0.80 . Catalogue matching, NTxPred2, mebipred and AnOxPePred then reduced the periodontitis branch to twelve explicit 7–9-aa peptides. All twelve docked to human AChE with best-run Vina affinities between -8.25 and -9.60 kcal/mol (Figure 2, Table 6). FLLHTTR, YLSLLQR and LLHPLRL contact the PAS. Over 100 ns, apo AChE and the ALLLHRC, FLLHTTR and YLSLLQR complexes remain globular; FLLHTTR and YLSLLQR become more compact than apo, and FLLHTTR keeps the densest intermolecular hydrogen-bond net (Table 7, Figures 5–7). These objects define a computational ligand set for the same PAS that accelerates A β assembly [4,18]. They do not demonstrate transport, expression or disease causality.

How the two libraries differ

The healthy and periodontitis libraries are close on most UniDL4BioPep heads (Tables 1–3). BBB (BBP) rates differ by 0.12 percentage points (9.72% versus 9.60%), so a BBB-high call is not a periodontitis stamp. Periodontitis exceeds healthy by ≥ 1 percentage point only for quorum sensing (+1.39). Healthy is higher for antifungal (-2.50), anti-parasitic (-2.36),

antioxidant FRS (−1.95), antiviral (−1.89), antibacterial (−1.42), CPP (−1.24), anticancer-main (−1.22) and umami (−1.00). Antimicrobial is essentially tied (87.89% versus 87.69%). Those percentages are library-wide score rates. Without a row-wise overlap map, the twelve docking sequences cannot be shown to be absent from the healthy library, and the periodontitis label cannot be transferred to an individual peptide. Physicochemical descriptors in Table 5 characterise the docking ligands; they are not a disease contrast.

Serial agreement among UniDL4BioPep, NTxPred2, mebpred and AnOxPePred is a filter stack. ESM embeddings, composition features and task-specific training sets can share correlated errors [23–26]. Near-universal antimicrobial positivity in both libraries illustrates the point: a score above 0.80 ranks sequences, it does not establish BBB transport, neurotoxicity, metal binding or redox chemistry. The same predict-then-filter order used to mine microbiome peptide antibiotics [31] is therefore retained as triage. Catalogue matches against HOMD, eHOMD and oral metaproteomes support prior observation of a string [21–22,32–34]; they do not prove translation in PRJNA678453 samples [7,15–16].

A possible PAS-centred path from oral peptides to AD

AD pairs amyloid deposition with cholinergic failure [2–3]. Independently of catalysis, AChE accelerates A β fibril assembly through the PAS, and AChE–A β complexes are more neurotoxic than A β alone [4]. A hydrophobic PAS motif is sufficient to promote that chaperone activity [5]. PAS-directed ligands suppress AChE-induced aggregation in biochemical assays [19]. Accelerated MD places A β on the AChE surface as a nucleation centre [17]. A 1- μ s trajectory keeps A β at the PAS, with principal residence at residues 344–361 [18]. Periodontitis and *P. gingivalis* supply an exposure context: the organism and gingipains have been reported in AD brains, and repeated oral infection in mice produces neuroinflammation and A β -related changes [6,12–13]. Docking and 100-ns trajectories indicate that periodontitis-derived micropeptides can occupy that same PAS. Four linked steps outline a possible mechanism.

1. PAS recognition and gorge-entrance occupancy.

Best poses of the twelve peptides concentrate at the PAS and gorge mouth of human AChE (PDB 4EY6; Figures 2–4). FLLHTTR anchors the canonical PAS residues Asp74, Tyr72 and His287 (best-run -9.60 kcal/mol; Figure 3C). YLSLLQR contacts PAS (Tyr72, Thr75) and the catalytic entrance (three-run mean -9.44 ± 0.09 kcal/mol; Figure 4L). LLHPLRL spans the PAS gatekeepers Trp286/Tyr341 to catalytic His447 (Figure 4I). HLLTLKKHV reaches Tyr72 and Phe346 in the 344–361 residence zone reported for A β . That geometry is the PAS Inestrosa identified as the pro-fibrillar site and that Atanasova occupied with A β . The aromatic gorge that joins the triad to the PAS was mapped on *Torpedo* AChE [20,36]. Vina ranks poses inside a fixed grid; the scores are not experimental free energies [27–28].

2. A surface-bound enzyme–peptide complex.

Over 100 ns the enzyme remains globular (RMSD 0.16–0.19 nm, Rg 2.30–2.31 nm, helix $\sim 33\%$ / sheet $\sim 17\%$; Figures 5–7). FLLHTTR and YLSLLQR complexes become more compact than apo in late RMSD (0.1640 and 0.1625 nm versus 0.1897 nm), so the peptide stays on the surface and stiffens the fold. Intermolecular hydrogen bonds persist: FLLHTTR holds a dense polar net throughout (7.03 ± 1.28 ; Figure 6F), YLSLLQR averages 2.93 ± 1.14 , and ALLLHRC retains seven contact pairs after early rearrangement. Peptide self-fit RMSD is lowest for YLSLLQR (0.0911 nm) and highest for ALLLHRC (0.2518 nm). Centre-of-mass RDF peaks lie at 1.22–1.80 nm, consistent with surface residence rather than bulk solvent. Lushchekina and Atanasova described a surface-bound, non-dissociating AChE–A β complex; the same occupancy pattern appears here for oral micropeptides, on a 100-ns rather than 1- μ s window.

3. Restricted acetylcholine access.

The PAS sits at the mouth of the 20-Å gorge that feeds the catalytic triad [3,20]. Physical occupancy of Asp74/Tyr72/Trp286/Tyr341 can hinder substrate entry even while the catalytic core remains folded (low RMSF in panels B of Figures 5–7). A pose that docks to the PAS therefore intersects the cholinergic axis of AD: reduced access of acetylcholine at synapses already depleted after basal-forebrain loss.

4. Pathological chaperone activity.

Because the PAS is a documented pro-fibrillar site [4–5], a heterologous peptide that remains there can lower the nucleation barrier for endogenous A β . FLLHTTR supplies a persistent polar net on the PAS (Figure 6F) that matches its docking pose (Figure 3C). YLSLLQR buries surface (SASA 209.71 versus 212.25 nm²; Figure 7C) and is the most rigid bound peptide. Folded AChE then presents a peptide-coated PAS on which A β oligomers can co-assemble, which is Lushchekina's nucleation-centre picture applied to an oral ligand. AChE–A β assemblies are already more synaptotoxic than free A β [4]. A bacterial micropeptide on the same site is a possible route to hybrid nuclei.

From the oral cavity to cortical AChE

Chronic periodontitis can move *P. gingivalis* products into blood through a disrupted epithelium, gingipains and outer-membrane vesicles [8–9]. Systemic cytokines and proteases increase BBB leakiness [6]. Short, leucine-rich, cationic peptides that scored BBB-high—a label almost as common in the healthy library—could in principle reach interstitial fluid [12,35]. PAS docking then supplies a landing site on an enzyme that is both a cholinergic hydrolase and an amyloid chaperone. Meta-analyses link periodontal disease to cognitive disorders, although effect sizes move with case definitions [10], and periodontitis tracked later decline in an AD cohort [11]. A two-sample Mendelian randomization analysis did not support a genetic causal effect of periodontal disease on AD [14]. The twelve sequences are treated as pathogenic candidates because they occupy the experimentally mapped A β -binding PAS and

remain bound for 100 ns. Complex RMSD that stays at or below apo is local adaptation of a folded enzyme, not evidence that the peptide is harmless or that it is proven to seed amyloid in tissue.

The calculation therefore connects three literatures that are usually kept apart: oral smORF peptide space [15,31], the PAS chaperone activity of AChE [4,18], and the periodontitis–AD exposure context [12–13]. What it supports is a bounded structural hypothesis. What it does not replace is measured expression, transport, affinity or an A β -aggregation assay.

Conclusions

Twelve 7–9-aa peptides from a periodontitis-labelled oral smORF library dock to human AChE. FLLHTTR, YLSLLQR and ALLHRC remain on the surface for 100 ns without unfolding the enzyme. FLLHTTR forms the densest PAS hydrogen-bond net; YLSLLQR is the only complex that buries solvent-accessible surface. Healthy and periodontitis libraries differ little on BBB-high rate; quorum sensing is the only head at which periodontitis is higher by ≥ 1 percentage point. Read against the amyloid cascade [2], the cholinergic hypothesis [3], and the PAS chaperone work of Inestrosa, Lushchekina and Atanasova, the calculations support a possible mechanism in which oral pathogenic peptides occupy AChE, hinder acetylcholine access, and co-nucleate A β on the same PAS.

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